

2(a)	secondary (structure) ;	1
2(b)	any one from: <i>idea that</i> area to be, hydrolysed / AW, could be between same amino acids ; A ref. to (always), hydrolyses / breaks down / acts on, peptide bonds suggestion that active site has (some) flexibility for, hydrolysing / binding to / AW, similar substrates ; (proteins / substrates) similar shape(s) so fit active site ; active site and, substrates / proteins, (still) have complementary shapes ; R 'complementary to the antigen or the shape' AVP ; e.g. (large enzyme / enzyme complex, with) more than one active site	1

Question	Answer	Marks
2(c)	any three from: fibrous ; A not globular I structural insoluble / does not dissolve / does not dissolve easily ; (high proportion of) disulfide bonds / disulfide bridges / covalent bonds (because of cysteines) ; high proportion of / many / AW, hydrogen bonds (between polypeptides) ; high proportion of / many, hydrophobic interactions ; A many, amino acids with hydrophobic R-groups / hydrophobic R-Groups tight / close, packing, of, chains / polypeptides / molecules ; I triple helix AVP ; e.g. arranged so peptide bonds are not accessible parts of molecule needing to bind to active site not accessible many amino acids with small R-groups / small amino acids I details specific to collagen e.g. gly every third amino acid amino acids	3

Question	Answer			Marks
2(d)	keratinase	temperature range at 60% relative activity / °C	pH range at 60% relative activity	4
	K12	41 – 63	6.8 / 6.9 – 8.4 ;	
	A22	36 / 36.5 – 57 ; A 57.1 / 57.2	6.0 – 9.0	
	P3	29 – 56	5.3 – 7.5	
	P3 ;			
	A22 ;			

Question	Answer	Marks
2(e)	<i>must have pH <u>and</u> °C <u>and</u> %, at least once max 2 if only Fig. 2.1 or only Fig. 2.2 used for the response.</i> <i>any three from:</i> <i>advantage from Fig. 2.1 (temperature)</i> 1 A22 <u>and</u> K12, active / remove stains, at a wide range of temperature or stated (active) temperature range of, A22 is 20.5 / 21 °C or K12 is 22 °C or use of Fig. 2.1 to give % activity and stated temperature of, good / above 60%, activity at, medium / hotter, temperature washes for, A22 / K12 ; or K12 active at low temperature (washes) ; A data e.g. 37–40% activity at 20 °C / 41–43% activity at 30 °C <i>disadvantage from Fig. 2.1 (temperature)</i> 2 low activity / poor stain removal, at 30 °C / 70 °C / 80 °C, for A22 or at 20 °C / 30 °C / 70 °C, for K12 ; A data extracted from Fig. 2.1 <i>advantage from Fig. 2.2 (pH) – activity in alkaline conditions</i> 3 A22, greatest activity / most effective / AW, pH 7.5 – 8.0 or (at least 60%) activity between pH 7.5 – 9.0 ; A range of pH 6.0 – 9.0 but R if stated pH 6.0–7.0 is alkaline or K12, active between pH 7.5 – 8.0 / 8.5 or optimum activity at pH 8.0 or data for (good) activity at stated pH values from Fig. 2.2 for either A22 or K12 ; <i>disadvantage from Fig. 2.2 (pH)</i> 4 use of Fig. 2.3 to show a decrease in activity at higher pH values ;	3

Question	Answer	Marks
2(e)	5 any wash feature in context of mp 1–4 ; e.g. have (some) activity at low temperatures so, more economic /less energy used (to heat water) A ora active at high temperatures but, more costly / needs (more) energy can wash at high(er) temperature so, can sanitise clothes / can kill insect pests / can kill pathogens / shorter time to remove stains comparison of K12 and A22 for washes e.g. K12 more, thermostable / stable in (washes at) higher temperatures K12 has higher optimum temperature so more, costly / energy used K12 only useful at lower alkaline pH A22 has wide(r) working range of pH A22 more tolerant to / stable at, higher pH	